Social media analysis Documentation

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Tools

I used Excel Power query And PowerBI Desktop, DAX

Source file

social_media_raw_v3_10k.csv

Data

social media posts, Jan 2024 to 19 Oct 2026

Output

Clean star schema + 5-page dashboard

1. Project overview Objective.I clean a Raw social media dataset.Madel it is Star schema, I build PowerBI dashboards 2 pages,how posts perform across platforms, countries, audiences and posting times.

2.Source Data

The raw file has 10,000 rows and 31 columns (9,500 unique posts after removing duplicates). It covers 6 platforms, 13 countries and 4,740 users. Each row is one post.

Post (10)	post_id, post_date, post_time, post_type, content_category, video_length_sec, platform, device, moderation_status, scrape_timestamp
User (8)	User_id, username, account_type, verified, follower_count, account_created_date, country, language
Engagement (8)	views, reach, likes, comments, shares, saves, followers_gained, engagement_rate
Sentiment (2)	sentiment_label, sentiment_score (-1 to 1)
Sponsorship (3)	is_sponsored, brand_name, campaign_id

3. Data cleaning log

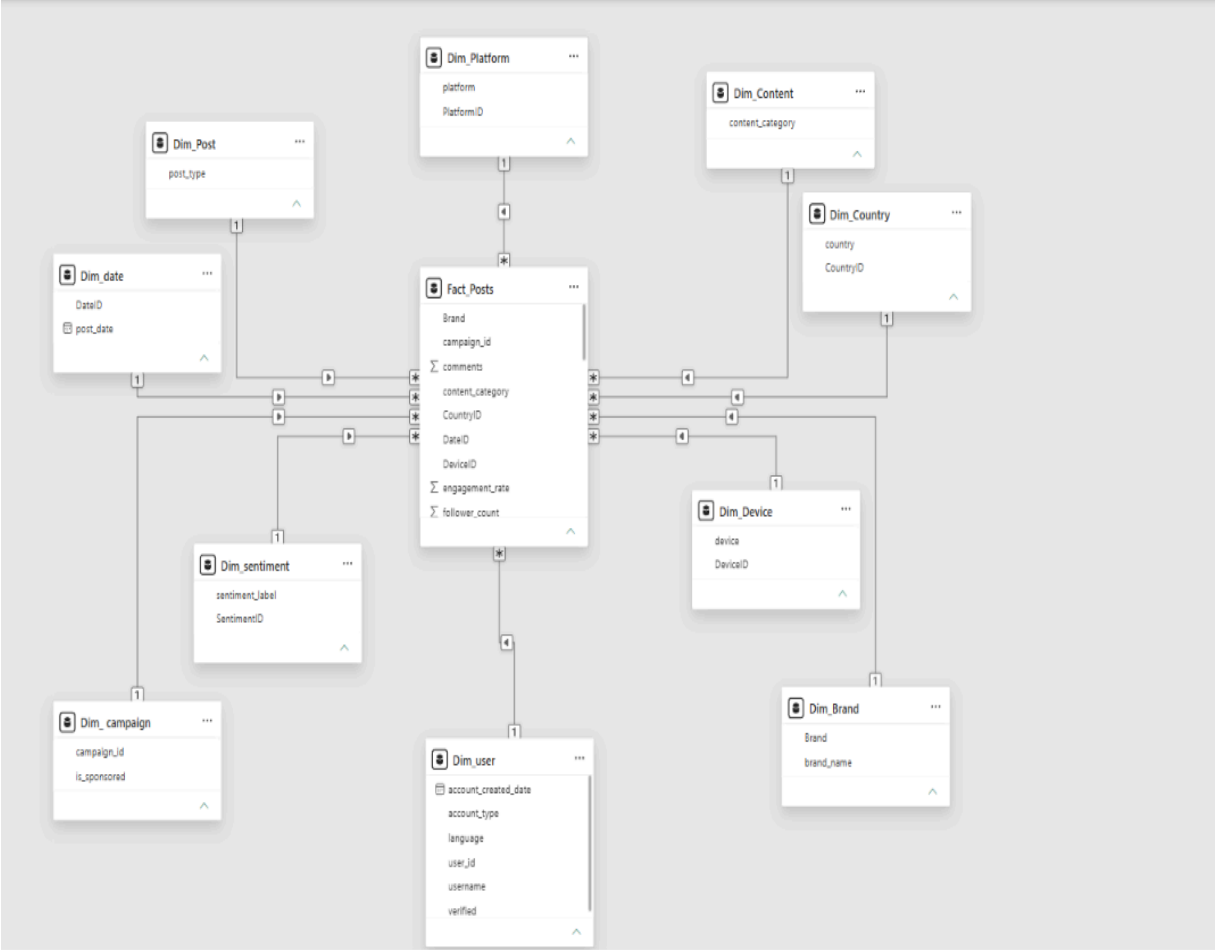
Issue	Column(s)	Found in raw data	Fix applied	Result

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Duplicate Rows	Post_Id	500 duplicate rows	Remove duplicates on Post Id	10,000 to 9,500 rows
Upper / lower case	country, content_category	39 country values, 45 category values	Capitalize each word	13 countries, 15 categories
Missing values	sentiment_score; device	616 blanks; 486 blanks	Fill score with median (0.15); fill device with "Unknown"	0 blanks
Inconsistent values	platform; sentiment_label	15 platform spellings (IG, Insta, X, FB...); 9 sentiment labels (Pos, Good, Bad...)	Find replace tool	6 platforms; 3 sentiments
Extra spaces	username, post_type, brand_name	811, 830 and 1,187 rows (leading, trailing, double spaces)	Trim	Clean text

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4. Data model (star schema)



The model has one fact table, Fact_Posts (one row per post), joined to 10 dimension tables by many-to-one relationships with single-direction filtering. A star schema keeps the data small and makes DAX measures simple.

Dimension tables and join keys: Dim_Platform (PlatformID, 6 rows), Dim_Country (CountryID, 13), Dim_date (DateID, 1,023), Dim_sentiment (SentimentID, 3), Dim_Brand (Brand, 25), Dim_Device (DeviceID, 4), Dim_Post (post_type, 13), Dim_Content (content_category, 15), Dim_user (user_id, 4,740) and Dim_ campaign (campaign_id, 792).

5.Measures (DAX)

Engagement Rate is calculated from totals, not by averaging the engagement_rate column, so big posts weigh more than small ones.

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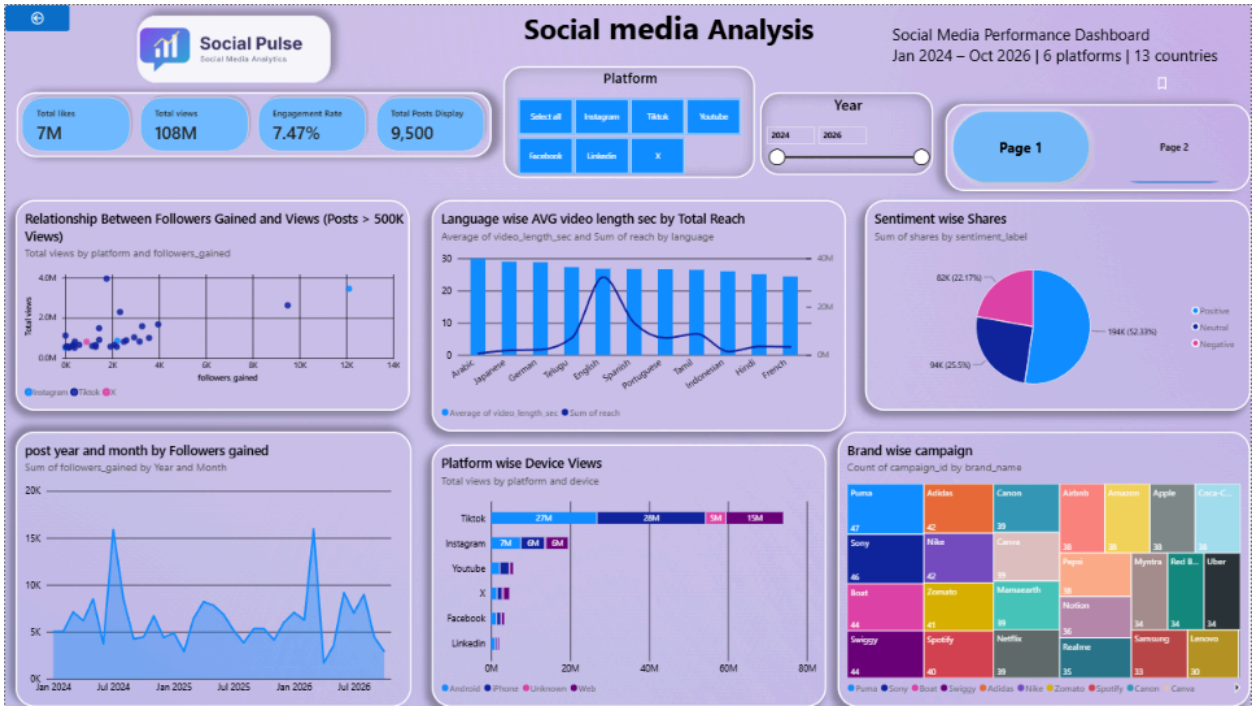
```
AVG YT Video length = CALCULATE(AVERAGE(Fact_Posts[video_length_sec]),Dim_Platform[platform] =
"YouTube")
Engagement Rate =
DIVIDE(
    SUM(Fact_Posts[likes]) + SUM(Fact_Posts[comments]) + SUM(Fact_Posts[shares]) + SUM(Fact_Posts[saves]),
    SUM(Fact_Posts[views])
)
Maximum video length = MAX(Fact_Posts[video_length_sec])
Total Campaign = DISTINCTCOUNT(Fact_Posts[campaign_id])
Total followers gained = SUM(Fact_Posts[followers_gained])
Total likes = SUM(Fact_Posts[likes])
Total posts = COUNT(Fact_Posts[post_id])
Total Reach = SUM(Fact_Posts[reach])
Total views = SUM(Fact_Posts[views])
```

6. Dashboard walkthrough

Every page has a Platform slicer, a year slicer, and Four KPI cards: Total Likes(7M),Total views(108M), Engagement Rate (7.74%) and Total Posts (9,500).



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Page	Main visuals	Question answered
Overview	KPI cards (posts, views, reach, engagement rate)	Overall account performance is positive
Platform	Average and Total views, posts, engagement rate by platform	Tik tok takes most reached
Content	Post type and category: posts and engagement rate	Gaming holds the top position at 10% and most verified people engage in lifestyle.
Audience	Country map and bar chart, account type, verified, follower size	India in top with 33M views And most verified people followed by news category.
Timing and sponsored	Posts by hour, weekday, sponsored share by account type	When posting at 6-9pm it is good reach to upload that time.

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7.Key Insights

Platform and follower count explain performance far better than sentiment or posting hour.

TikTok leads: about 38.6K views per post against 1.8K for Facebook, and 68% of total reach.

Two countries give half of the views: India 30.8% and the United States 19.3% of 108.5M views.

More followers means more views: the greater than 500k gained to Tiktok, instagram and X but most followers gained Instagram 12.1K and then Tiktok 9.5k gained followers.

Sentiment does not change engagement: Positive sentiment has shared (194k) 52.23% shares.

Business, Brand and Creator accounts sponsor more: about 16% of their posts, against about 4% for Personal and Public Figure accounts. Overall, 90% of posts have no brand.

Evenings (6 PM to 9 PM) are the busiest posting time, and Gaming has the best engagement rate at about 10%.